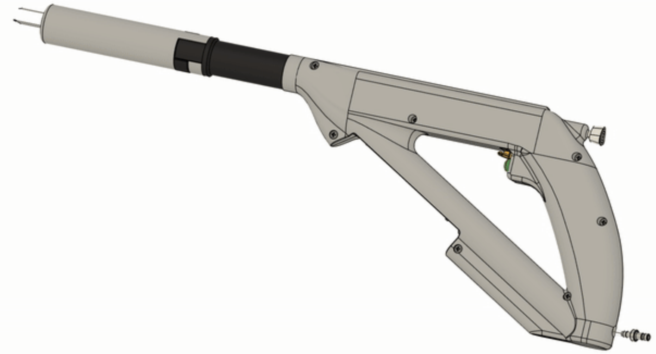
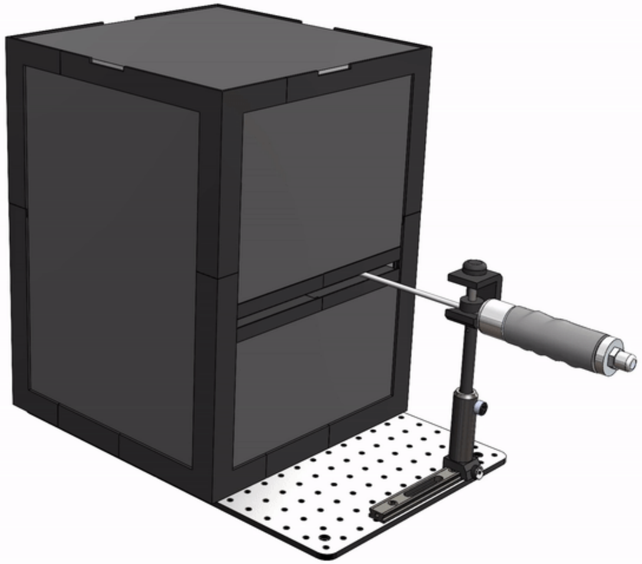
NOTE: There are embedded GIFs that only works in the PowerPoint file



CAD Portfolio Highlights v2

Full Portfolio: https://github.com/jchen1231/CAD_Portfolio



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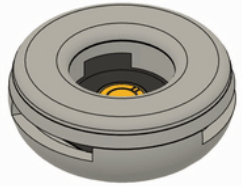
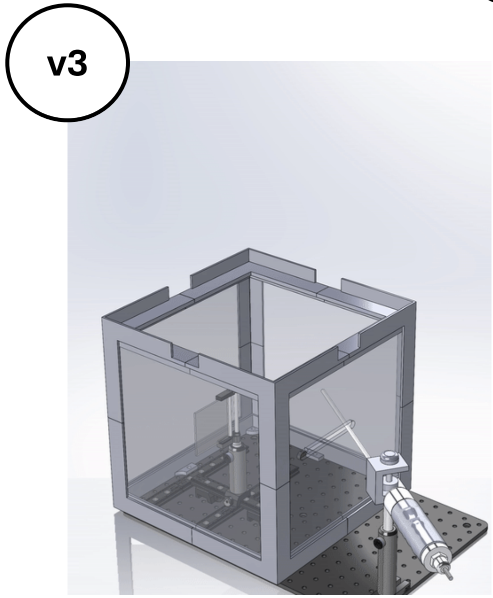
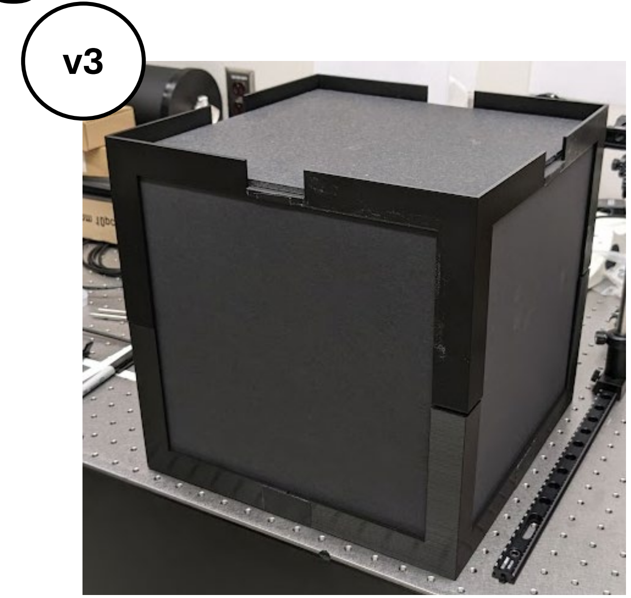


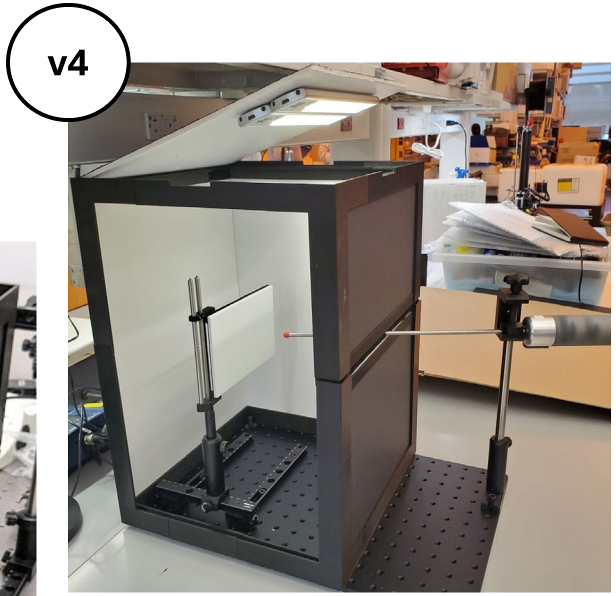
Project 1: Portable Testing Chamber

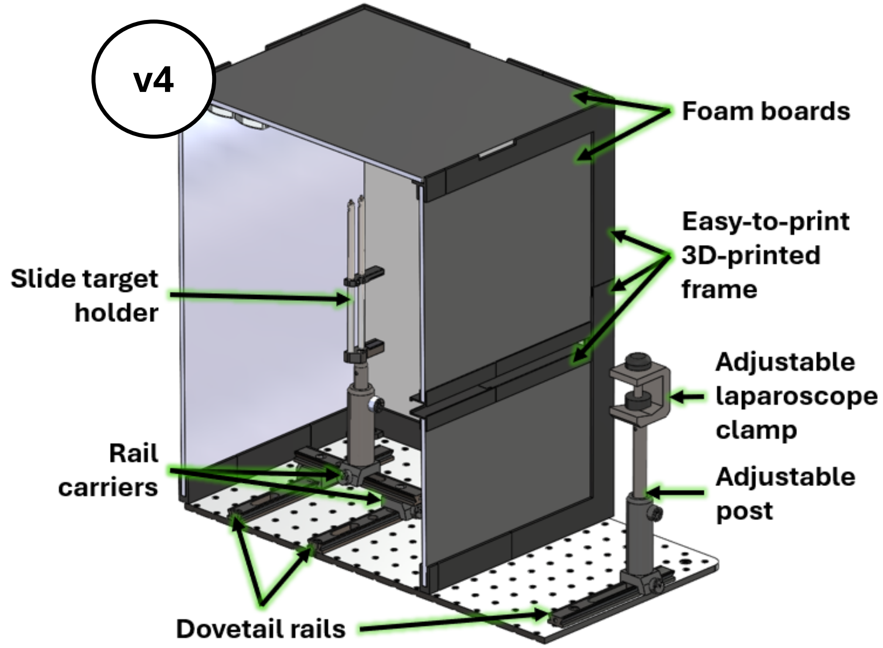
Develop an accessible and versatile bench testing setup for laparoscope and camera characterization, especially in low-resource or field-deployable settings.

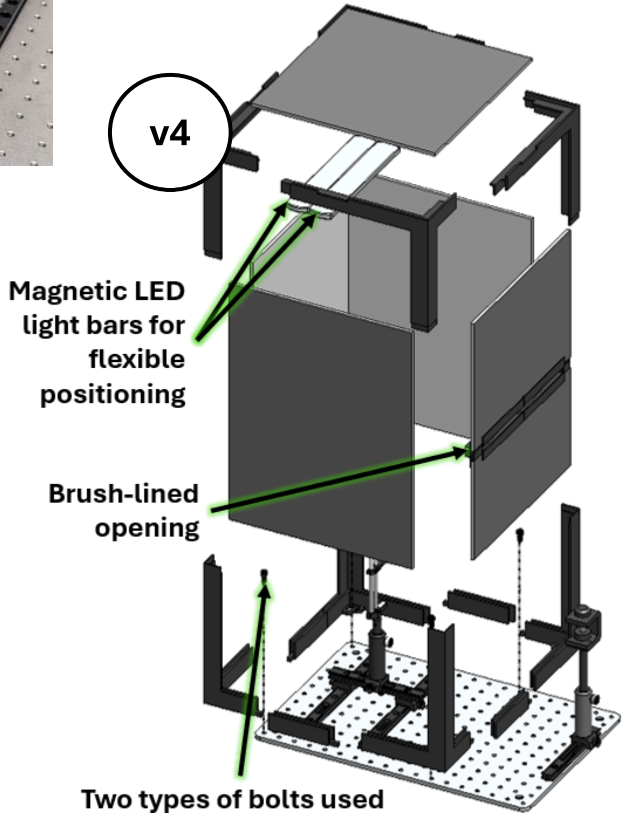
- We ensured that our desired camera working distances (30–100 mm) were accommodated, and the PTC remained compatible with our image analysis targets, enabling consistent evaluation and effective optimization after each design iteration.
- The brush-lined opening makes the PTC compatible with various laparoscopes, including our lab's 5 mm, 10 mm, and 30-degree models.
- The enclosure is primarily made of PLA, foam, and paper, costing less than 25% of alternatives such as Thorlabs' XE25C7/M.
- We split the frame into multiple components to make it printable on hobbyist 3D printers such as the Ender 3 Pro.
- We designed disposable compact plastic nuts to be tapped by bolts, as the desired dimensions were not commercially available.





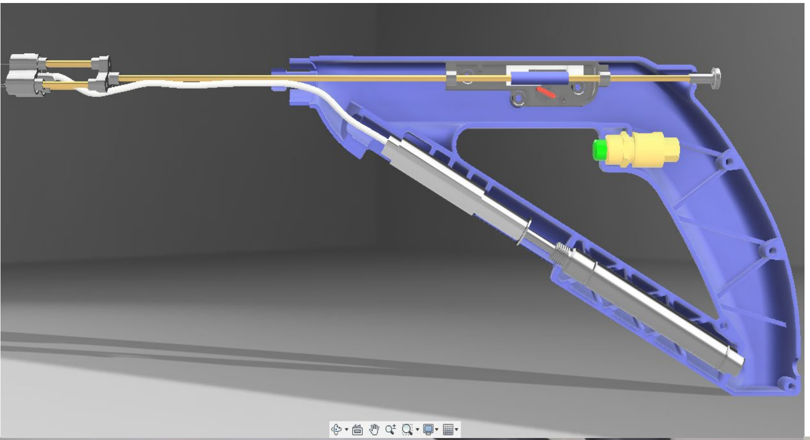






Project 2: Auto-injector

Legacy



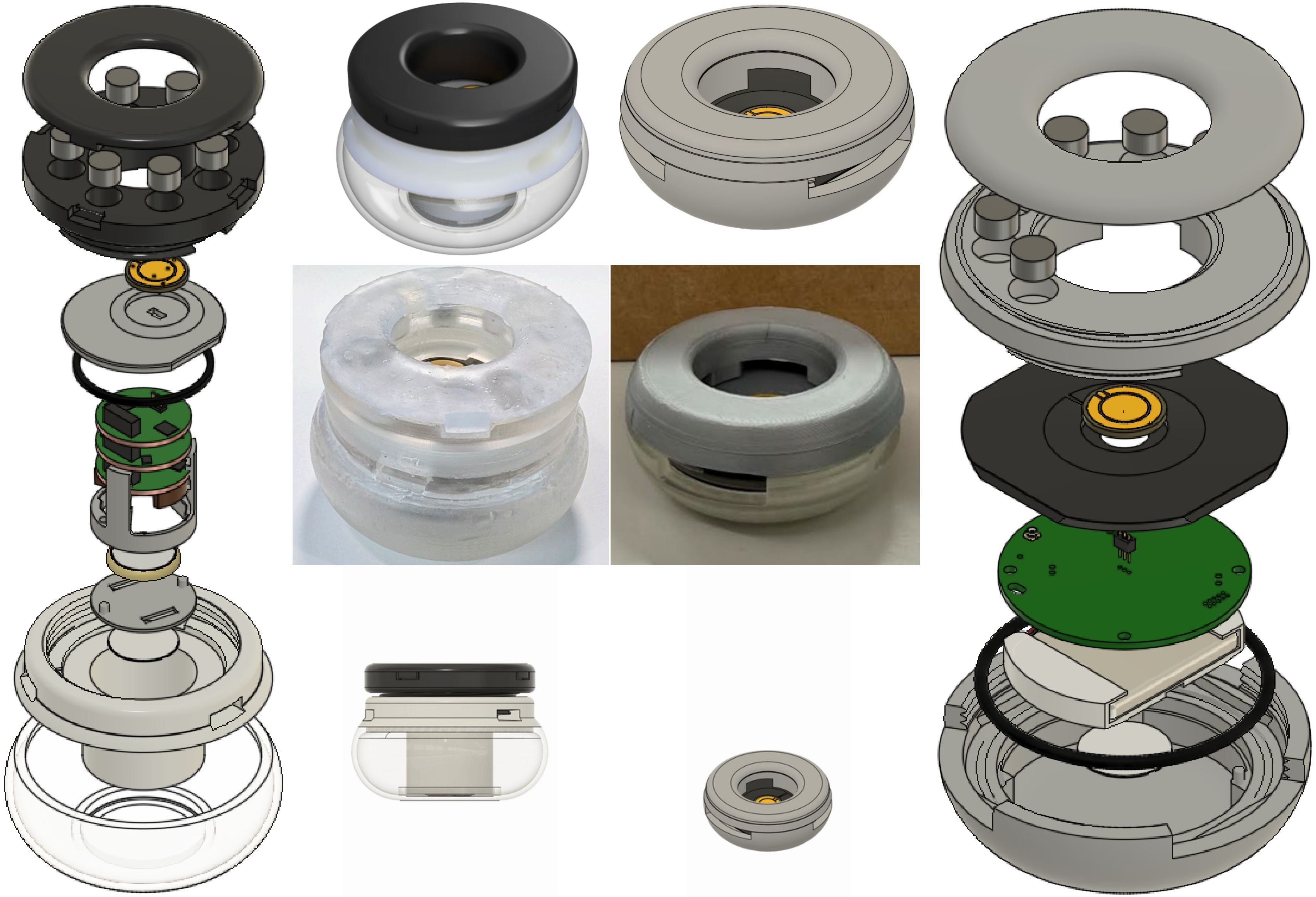
Updated

Develop an easy-to-use, low-cost triple-needle injector that delivers three simultaneous deposits of a tumor-ablating therapeutic, creating a larger area of tissue necrosis with a single injection.

- Designed and printed detachable and speculum-compatible needle covers.
- The distal triple-needle assembly is detachable through an added threaded insert; the long rod and springs can now remain in place.
- The retraction mechanism now functions reliably after minimizing friction, refining latch designs, and integrating stronger springs.
- The retraction housing is fully modular and is held in place using alignment holes/ribs, friction, and pressure fits.
- Adding a trigger button latch removes the need to constantly hold down the button.
- Maintained a housing design compatible with injection molding.
- Reduced the number of screws and other pneumatic components.
- Air canister is compatible with both bike pumps and air compressors.

Project 3: Bobblestat Biosensor

Revision 1 **Revision 2**



Develop a watertight, modular, and buoyant biosensor that integrates potentiostat PCB and swappable electrodes.

- Switched to POGO pins, which offer greater longevity and reduce assembly complexity but require more precise alignment.
- Reduced the number of weight slots to only what is needed for orientation.
- Converted snap-fit lid to screw-on lid to improve fit and durability.
- O-ring groove location moved from the electrode substrate to the housing side, and it now features a dovetail cross-section to improve retention and reduce disposables.
- Modified the substrate to be squarer, providing a more stable fit than the previous circular outline.
- Enlarged and realigned the degassing slots to facilitate air bubble escape.
- Added an alignment groove to the substrate to aid in electrode orientation.
- This design ensures the antenna always faces upward when floating, providing optimal signal quality.